

Certifying the optimality of an unsupervised-metric prompt

What can be proved about p^* with no ground truth

(working draft)

Abstract

We certify optimality of a prompt defining a single latent evaluation metric M_i , with no labels and no stronger-model anchor. The objective is a label-free *recovery* R_i : the TVD mutual information between the metric’s held-out verdicts and a reconstruction obtained by articulating it as a rubric and re-executing it on the same held-out items. The certificate is a *two-sided bracket*. The lower side is free and tight: $R_i \leq T_i = I(M_i; X)$ by data processing, and this transmission ceiling is globally attained by the metric’s own posterior readout, so recovery’s optimum over *all* prompts is exactly T_i —but T_i is the ceiling on the *operationalized* metric m_ω , hence only the **floor** on the ideal meaning-defined metric M_i^* . The upper side—bounding M_i^* from above—is the harder job and is necessarily unsupervised (there is **no** aggregate label Y): we give the $\mathcal{B}_E$ census (a Heaps/Good–Turing go/no-go on whether the reachable criterion space can be covered), an R -weighted cross-source consensus extrapolation $H(c_\infty)$, and a prompt-free spectral decoder-capacity ceiling OPT' . Every quantity is *per-metric*, and every $\mathcal{B}_E$ claim is *process-relative* to a specific non-IID set of criterion generators. Along the way: a universal cap, a within-class submodular certificate with an exact conditional theorem, a finite-set treatment of presentation form, and a discovery scaling law—each with its assumptions and looseness stated.

1 Setup and objective

Unit of analysis: one metric (read this first). Everything below scopes to a *single* latent evaluation metric M_i —one per R2 cluster (e.g. “novelty of contribution” in peer review). A task has j such metrics $M_1, \dots, M_j$; the observed outcome $Y = f(M_1, \dots, M_j)$ (e.g. accept/reject) is their *aggregate* and is **not** the target—the loop below is anchor- and label-free (**no** Y). M_i is itself unlabeled; we recover it from its *own* (item, verdict) pattern. Every symbol ($R, T, \Omega, \mathcal{B}_E, \alpha, \gamma, A$) is per-metric ($R_i, T_i, \alpha_i, \dots$); repeat the analysis for each metric. This is load-bearing and recurs as a bug: at the *task* level (pooling every metric’s criteria) the criterion universe is always vast, so the Heaps exponent $\alpha \approx 0.9$ *everywhere* (non-discriminating). The structure—thin vs. thick, coverable vs. inexhaustible—lives only per-metric ($\alpha_i < 0.5$ for a simple metric, $\alpha_i \rightarrow 1$ for a complex one).

Items, behavior. Items $X_1, \dots, X_N$; index $I \sim \text{Unif}[N]$. A prompt p makes executor E a stochastic map item $\rightarrow$ verdict; binary behavior $s_p = (s_1, \dots, s_N)$, $s_i = \Pr(V=1 \mid X_i, p)$; K -ary $s_p \in (\Delta_K)^N$. Behavior space $\mathcal{C} = (\Delta_K)^N$ is convex, compact. The achievable set $\mathcal{B}_E = \{s_p : p \in \Sigma^*\}$ (for this metric) is countable, non-convex, known only through a sampling oracle.

Recovery (the objective). Split items into train X_{tr} and held-out X_{te} . Run p on X_{tr} to get $M_{\text{tr}} = m(X_{\text{tr}})$; a reconstructor reads $(X_{\text{tr}}, M_{\text{tr}})$ and emits a rubric $\hat{s}$; run both on the *same* held-out items for $M_{\text{te}} = m(X_{\text{te}})$ and $\widehat{M}_{\text{te}} = \text{exec}(\hat{s}, X_{\text{te}})$. Then

$$R_i(p) = I_{\text{TVD}}(M_{\text{te}}, \widehat{M}_{\text{te}}), \quad I_f(A; B) = D_f(P_{AB} \parallel P_A \otimes P_B).$$

R is label- and anchor-free; it is high only if p runs a generalizable rule, not memorized train features.

Transmission. $T(p) = I_f(I; V)$ under one behavior s_p . Distinguish in-sample consistency T_{tr} from the held-out $T_{\text{te}} = I_f(\widehat{M}_{\text{te}}; X_{\text{te}})$; only the latter bounds R (Prop. 2). T is not the objective—it is the convex quantity that caps R .

Operationalized vs. ideal metric. Write m_ω for the metric as actually operationalized by the executor on its criteria (what the verdicts realize), and M_i^* for the *ideal*, meaning-defined metric the cluster reaches for. These differ: m_ω is reachable and measured, whereas M_i^* is a *meaning* object—unreachable by maximizing recovery, since maximizing R finds decodability, never meaning. There is no label for M_i^* , and—again—no Y .

Reduction. p matters only through s_p , so $\text{OPT} := \sup_p R(p) = \sup_{s \in \mathcal{B}_E} F(s)$.

The two-sided bracket (the whole paper).

$$\underbrace{\widehat{R}_i(\hat{p})}_{\text{free lower bound}} \leq \text{OPT}_i = T_i(m_\omega) \leq \underbrace{I(M_i^*; X)}_{\text{the ideal}} \leq \underbrace{U_{\mathcal{B}_E}}_{\mathcal{B}_E \text{ census}}.$$

The lower half (through $T_i(m_\omega)$) is the recovery certificate of §3–§11; the crucial orientation is that $T_i(m_\omega)$ is the *floor* on the ideal M_i^* , never its ceiling. The upper half—bounding M_i^* from above via $\mathcal{B}_E$ —is §12, and is unsupervised end to end.

Definition 1 (Certificate). *For a found $\hat{p}$, an ε -global certificate is a verified $U \geq \text{OPT}$ with $U - \widehat{R}(\hat{p}) \leq \varepsilon$. The lower bound is free; the work is producing U .*

2 Two structural facts

Proposition 1 (Convexity of T). *$T(s) = I_f(I; V)$ is convex in s , so $\max_{s \in \mathcal{C}} T$ is attained at a deterministic per-item labeling.*

Proof. $P_{I,V}(i, v) = \frac{1}{N} p_i(v)$ and $P_I \otimes P_V$ are affine in s ; $(P, Q) \mapsto D_f(P \| Q)$ is jointly convex; convex $\circ$ affine is convex. A convex function on a compact convex set attains its maximum at an extreme point of $\mathcal{C}$ (Bauer)—a vertex, i.e. a deterministic labeling. $\square$

Proposition 2 (Data processing: $R \leq T_{\text{te}}$). *On the held-out split, $R = I_f(M_{\text{te}}; \widehat{M}_{\text{te}}) \leq I_f(M_{\text{te}}; X_{\text{te}}) = T_{\text{te}}(m)$.*

Proof. $\hat{s}$ is learned on the disjoint train split, so $\widehat{M}_{\text{te}}$ is produced from X_{te} by a channel independent of M_{te} given X_{te} . The f-divergence DPI ($D_f(KP \| KQ) \leq D_f(P \| Q)$) applied to that channel gives $I_f(M; \widehat{M}) \leq I_f(M; X)$. The inequality lives within one distribution: it does *not* hold against the train consistency T_{tr} . $\square$

Asymmetry. T is convex; R (a nonlinear learned map of s) is not known convex. Hence we cap R by T globally and optimize R only locally / within-class.

3 The recovery ceiling is global—and it is the floor on the ideal

Prop. 2 bounds R by T_{te} for the found prompt; more is true *globally*. Since $M \perp \widehat{M} \mid X$ (the reconstructor, trained on a disjoint split, never sees X_{te}), $M \rightarrow X \rightarrow \widehat{M}$ is a Markov chain for *every* prompt the reconstructor could emit, so the f-DPI gives

$$\text{OPT}_{\text{rec}} = \sup_{p \in \Sigma^*} R(p) = T(m_\omega) = I_f(m_\omega; X),$$

attained when the reconstruction outputs the *soft posterior* $\widehat{M} = \eta(X) := \Pr(M_\omega=1 \mid X) = \mathbb{E}[m_\omega \mid X]$, the Bayes-sufficient statistic ($I(m_\omega; \eta(X)) = I(m_\omega; X)$). Two consequences:

- **Both-readouts (continuous $\Pr(\text{YES})$) is *necessary* for tightness, not a nicety.** A *sam-pled* binary reconstruction is a conditionally-independent copy and gives $R < T$ strictly whenever the readout is stochastic given X ($H(M_\omega \mid X) > 0$). Only the soft posterior makes T attainable in principle, so that headroom $= T - R$ measures real articulability rather than an artifact of hard-rounding.
- **This is Rung 1 at the executor's real transmission,** not the vacuous $\text{cap}_f = \frac{1}{2}$: no prompt anywhere in $\Sigma^\star$ recovers more than $T(m_\omega)$, so “ $\hat{p}$ is ε -optimal for the metric” is exactly headroom $= T(m_\omega) - \widehat{R}(\hat{p}) \leq \varepsilon$. Recovery *dissolves* the class boundary: its effective class is $\Sigma^\star$ itself (free-gen can produce any prompt), so it never pays a within-/out-of-class tax.

Orientation—do not over-read T . $T(m_\omega)$ is the ceiling on recovering the *operationalized* m_ω , which equals the **floor** on the ideal $M_i^\star$: a richer criterion pool reaches more of $M_i^\star$, so $T(m_\omega) \leq I(M_i^\star; X)$. The ideal's *upper* bound is a different object (§12), approached unsupervised, with no Y . The three nested ceilings, then, are $R \leq T(m_\omega) \leq \text{cap}_f$ on the operationalized side, and $T(m_\omega) \leq I(M_i^\star; X)$ crossing to the ideal.

The gap is a sufficiency deficit; for recovery there is no $\mathcal{B}_E$ search. The chain rule sharpens the headroom to $T - R(\varphi) = I(m_\omega; X) - I(m_\omega; \varphi(X)) = I(m_\omega; X \mid \varphi(X)) \geq 0$, zero iff the readout $\varphi(X)$ is sufficient for m_ω . So the recovery value-surface on $\mathcal{B}_E$ is not a black box with hidden peaks: it has *one* known global maximum ($\eta(X)$) and falls off as a Bregman-type deficit. Hence for *known-target* recovery there is no census of $\mathcal{B}_E$ —estimate η with the strongest model and both-readouts, measure the deficit, certify via T ; the only residual search is over reconstructor prompts (a saturation test on achieved R). The $\mathcal{B}_E$ census of §12 is for the *other* regime—the ideal / unknown-target, where no single η exists and value is coverage, not a deficit from a known peak.

4 The universal cap (Rung 1)

Theorem 1 (Cap). *For every p , $R(p) \leq T(p) \leq \text{cap}_f := \sup_{s \in \mathcal{C}} I_f(I; V)$, with*

$$\text{cap}_{\text{TVD}} = 1 - \frac{1}{\min(N, K)} \quad (\text{binary: } \tfrac{1}{2}), \quad \text{cap}_{\text{Shannon}} = \log \min(N, K).$$

Proof. For a deterministic labeling $V = g(I)$ the item index collapses and $I_{\text{TVD}}(I; V) = 1 - \sum_v \Pr(V=v)^2$ (Gini impurity), maximized by a uniform histogram over $\min(N, K)$ values. Prop. 1 places the maximizer there; $R \leq T$ extends the bound to recovery. $\square$

Comments. (i) K , not N , sets the ceiling: a K -symbol verdict narrows the item by a fixed *fraction*; N is sample size, entering only via $\min(N, K)$. A binary readout caps every R at $\frac{1}{2}$. (ii) Use the cap as a *sanity check* ($\widehat{R} \leq \text{cap}_f$; detect readout compression), *not* a proximity KPI: $\text{cap}_f - \widehat{R}$ is valid but vacuous, as usually $\text{OPT} \ll \text{cap}_f$.

5 The certificate ladder

Rung 1. $U_1 = \text{cap}_f$: global over $\Sigma^\star$, assumption-free, loose. (For *recovery* it upgrades to the tight $T(m_\omega)$ of §3.)

Rung 2. Restrict to $S \subseteq \Omega$ for a discovered criterion set Ω ; $f(S) = R(C(S))$ is a set function, certified by submodular selection (§7). Tight, relative to Ω .

Rung 3. Best-in-set over a finite materialized prompt list, with bootstrap CIs.

Rung 4. Local: edit-stationarity / saturation at $\hat{p}$.

Only Rung 1 speaks to Σ^* ; Rungs 2–4 buy tightness by downscoping (validity vs. tightness).

6 Constructing Ω

Each criterion $e \in \Omega$ is individuated *behaviorally* by its per-item signal $X_e \in \{0,1\}^N$: $e \equiv e'$ iff $X_e = X_{e'}$, regardless of wording. Glue text carries no X_e and is not in Ω ; “green and round” is one unit if E emits a single verdict for it.

Discovery-to-Selection.

1. **Mine diffs.** Harvest the criteria each accepted optimizer mutation *added* across the whole lineage $\rightarrow$ pool R_{pool} .
2. **Orthogonalize.** Add e to Ω iff $I(X_e; X_\Omega)$ is low (new partition), else discard (paraphrase). TVD-MI has no chain rule, so this high-dimensional CMI is a *Shannon* proxy: the cross-entropy reduction of a surrogate classifier predicting X_e from X_Ω . Survivors are near-orthogonal $\Rightarrow$ the CI-given- M premise of §7 approximately holds.
3. **Canonicalize.** A fixed-template compiler $C(S)$ (section order Format, Semantics, Negative) makes $f(S) = R(C(S))$ a genuine set function, killing path-dependence.
4. **Select.** Brute-force for $|\Omega| \leq 15$ (exact, bypasses γ); else §7. Shannon builds Ω ; TVD certifies on it.

Remark 1 (The prompt is prose; Ω is its decomposition). *The optimized prompt $\hat{p}$ is natural-language prose (the object E reads and scores with); Ω is a separate decomposition of $\hat{p}$ into atomic criteria, performed by a strong model after optimization—it is not the prompt itself, and structured JSON is never forced into the executed prompt. Writing M for E ’s verdict on $\hat{p}$ and M_Ω for E ’s verdict on the all-criteria compilation $C(\Omega)$, the certificate measures two distinct pairwise quantities:*

$$I(M; M_\Omega) \quad (\text{decomposition gap}), \quad I(M_\Omega; M_S) \quad (\text{selection gap}),$$

each read against its own reliability ceiling T (two passes of the same prompt, §2). The direct $I(M; M_S)$ would conflate a lossy decomposition with a bad subset; the two-step chain separates them. These are reconstructibility measures, not correctness (§4.3).

7 Submodularity: a conditional theorem and a greedy guarantee

Model criterion i as X_i ; ideal recovery $\check{I}(S) = I(M; X_S)$.

Lemma 1 (Monotonicity). $\check{I}(S \cup \{i\}) - \check{I}(S) = I(M; X_i \mid X_S) \geq 0$.

Theorem 2 (Submodularity signature).

$$I(M; X_i \mid X_j) - I(M; X_i) = I(X_i; X_j \mid M) - I(X_i; X_j).$$

Redundancy ($I(X_i; X_j \mid M) < I(X_i; X_j)$) shrinks the marginal gain (submodular); synergy (XOR-like) grows it (super-modular). Conditional independence given $M \Rightarrow \check{I}$ is monotone submodular.

Proof. Both differences equal the symmetric interaction information $I(X_i; X_j; M) := I(X_i; X_j \mid M) - I(X_i; X_j)$, which is invariant under permuting its three arguments; conditioning on X_j instead of M writes the same quantity as $I(M; X_i \mid X_j) - I(M; X_i)$. Equating the two expressions gives the identity. $\square$

Corollary 1 (Greedy guarantee). *With submodularity ratio γ and curvature α , $f(S_{\text{greedy}}) \geq \frac{1}{\alpha}(1 - e^{-\alpha\gamma}) \text{OPT}_\Omega$ (Bian et al.; Das–Kempe $1 - e^{-\gamma}$); $\gamma = 1$ gives $1 - 1/e$.*

Comments. γ is measured on Ω as discovered (mining with an “independent criteria” instruction would pre-bias $\gamma \rightarrow 1$). For $|\Omega| \leq 15$ brute force is exact and bypasses γ . We drop the spectral $\gamma \geq \lambda_{\min}$ bound: it presumes a linear-regression value function, false for an attention-mixing E . The realized $R(S) = I(M; \widehat{M}_S) \leq \widehat{I}(S)$ passes through an executor that compresses X_S into a single verdict, which can make R non-monotone—then use non-monotone double-greedy (BFNS: $\frac{1}{3}$ deterministic, $\frac{1}{2}$ randomized).

Remark 2 (Validity: side-channels, not submodularity). *Submodularity (§7) is the combinatorial axis (diminishing returns). Validity is orthogonal: is I causal/semantic, or a side-channel (length, formatting, a surface marker the executor latches onto)? I_{TVD} is correlation, not causal, so a criterion riding a confound inflates R spuriously. We block this by construction, not by adjustment: I is a v -information through the constrained class $V = \{\text{criterion} \rightarrow E\}$, so a causally-semantic criterion class excludes side-channels with no causal model of the data (Xu et al. 2020)—ignorability as a property of the channel class, not the confounder graph. PRUNE-help (removing e raises $R(C(\Omega))$) is the leakage alarm. Cleaning channels removes the spurious non-monotonicity of §7, so near-submodularity emerges from validity rather than being assumed—we never prove γ standalone.*

8 The form axis

Form φ (few-shot count, persona, format, order) carries no X_e ; it modulates executor fidelity. Certify the product

$$\max_{\varphi \in \Phi} \left[\max_{S \subseteq \Omega} R(S, \varphi) \right],$$

inner submodular-certified, outer best-in-set over a materialized Φ (Rung 3), no separability assumed. The canonical compiler fixes one order so f is a set function; its abstraction error equals the executor’s *order-sensitivity*, checked by a permutation test: with σ_{subset}^2 the variance of R across subsets and σ_{perm}^2 across orderings of one subset, the abstraction holds iff $\sigma_{\text{subset}}^2 \gg \sigma_{\text{perm}}^2$ (order is a residual of size $\sqrt{\sigma_{\text{perm}}^2}$). Folding order in would blow $2^{|\Omega|}$ subsets up to $\sum_k k! \binom{|\Omega|}{k}$ sequences.

9 Discovery scaling law: $\Omega \rightarrow \infty$

OPT_Ω is monotone non-decreasing in Ω and bounded by cap_f , hence converges $\text{OPT}_{\Omega_t} \rightarrow \text{OPT}_\infty$. A rate follows from universe curvature (high-impact criteria found early) and a Heaps/Zipf discovery curve; subsampled submodular maximization ($\sim (n/k) \log(1/\varepsilon)$ random elements retain $1 - 1/e - \varepsilon$ of OPT) converts discovery count into an approximation guarantee.

Magic-word defense (missing impact, not mass). Two prongs: (i) a submodular tail-bound,

$$\max_{e \notin \Omega} \Delta(e \mid \Omega) \leq \min_i [\text{greedy gain}]_i \quad (\gamma \approx 1 \text{ in the tail}),$$

where the leave-one-out minimum is a tighter diagnostic, not a guaranteed bound; and (ii) adversarial saturation, $I(X_{\text{probe}}; M \mid X_\Omega) \approx 0$ for rare/OOD probes (no submodularity needed). Halt

mining when both fire. (Good–Turing missing-*mass* cannot bound impact and is dropped here; it returns in §12 as a *coverage* certificate on the ideal, a different use.)

10 The verifiable corner: $E = \text{compiler}$

If M is code-generated a perfectly fitting rule exists ($R = 1$ reachable); if E is a compiler the within-item entropy is 0 ($T \equiv 1$). Then $R \leq T = 1$ is vacuous and $A = 1 - R$ is *pure learnability*. This gives calibration gates: a planted code metric must recover $R \rightarrow \text{cap}$, and planted submodular vs. synergistic constructions must recover γ . Restricting the executor class decomposes the same R :

$$\underbrace{R(\text{code, compiler})}_{\text{Verifiable}} \leq \underbrace{R(\text{rubric, LLM})}_{+ \text{Articulable}} \leq \underbrace{R(\text{dense model})}_{+ \text{Taste} = C}.$$

11 The articulation gap $A = T - R$

$A(m) = T_{\text{te}}(m) - R(m) \geq 0$ (same held-out split, Prop. 2). It measures the metric, not the prompt, and bundles three things a scalar cannot separate: tacit content, non-generalizing (overfit) spread, and executor-limited content. So A upper-bounds the tacit residual; the executor axis and the held-out split separate the parts, and the overfit part is a generalization gap $O(\sqrt{d/N_{\text{te}}})$ in the reconstruction family’s complexity d .

12 Upper-bounding the ideal M_i^* —the $\mathcal{B}_E$ census

The lower half brackets the *operationalized* m_ω . Bounding the *ideal* M_i^* from above requires saying something about $\mathcal{B}_E$ *beyond* the current pool Ω , and it must be unsupervised—**there is no Y** .

$$\underbrace{R(\Omega)}_{\text{floor: a richer pool does at least this well}} \leq \text{measurability of } M_i^* \leq \underbrace{?}_{\text{from } \mathcal{B}_E, \text{ beyond } \Omega}.$$

Assumption-free $\Rightarrow$ only the loose cap. With no structure the sup over reachable behaviors is the information cap (§4); when the reachable space is inexhaustible ($\alpha_i \approx 1$) the raw- $\mathcal{B}_E$ route *degenerates* to exactly that cap ($I(\text{all-of-}\mathcal{B}_E; X) \rightarrow H(X)$). Sampling $\mathcal{B}_E$ harder cannot tighten it—the raw- $\mathcal{B}_E$ -extent route is a dead end for a tight ceiling.

ALPHA-PROBE: the run-first go/no-go (per metric). Estimate the Heaps exponent α_i of M_i ’s reachable criterion space, with Good–Turing/Chao1 coverage and an assumption-free Berend–Kontorovich lower bound on covered mass, $C_{\text{lo}} = 1 - (f_1/N + \sqrt{\log(1/\delta)/N})$ ($N \approx 1200$ for a 0.95 certificate). Then $\alpha_i \ll 1 \Rightarrow \mathcal{B}_E$ (for this metric) is low-dimensional: cover it, get a tight U_i ; $\alpha_i \approx 1 \Rightarrow$ inexhaustible: enumeration cannot tighten, so the upper bound falls back to the direct DPI ceiling T_i (still valid, just not tightened by coverage). Run this *once per metric*; the task-level $\alpha \approx 0.9$ is the wrong level.

Support-completeness is not certifiable. No independence among the discovery lists can certify that their union support equals $\mathcal{B}_E$: a criterion in no family’s support has zero capture probability under every list (the DPI-guarded zero-mass tail), so the gap is a *shared-support ceiling*, not a correlation—more diverse lists narrow it but never close it. Three assumptions *could* close it, each trading the untestable support claim for another: **(a) positivity** (every articulable criterion has

proposal probability $\geq p_{\min} > 0$) makes coverage-of-support = coverage-of- $\mathcal{B}_E$; **(b) Lipschitz-impact** (value L -smooth in a criterion embedding) does not recover unseen species but *bounds their impact*—turning “unseen support” into “bounded residual impact,” the target we actually care about; **(c) complexity** (finite VC / Heaps $\alpha < 1$), which we distrust (non-Lipschitz extrapolation). The only lever that raises the LM-support ceiling itself is a *non-LM* (human/expert) elicitation list; if it too surfaces nothing new, that is the strongest available—still non-certifying—evidence of exhaustion.

Counting vs. value: the VALUE-CENSUS. ALPHA-PROBE puts the *counting* measure on M_i ’s criteria (how fast new ones appear); it is silent on *value*. A NO-GO ($\alpha_i \approx 1$) says M_i has inexhaustibly many *expressible* criteria, not that adding them keeps recovering M_i . Swap in the *value* measure—each criterion’s contribution to recovering M_i (its §3 recovery, never the aggregate Y)—and re-run the same machinery: the value exponent $\alpha_{V,i}$ can be small even when $\alpha_i \approx 1$. The gap $\alpha_i \gg \alpha_{V,i}$ is the per-metric signal “many expressible criteria, few actually pin down M_i ”: T_i is reachable but not by enumeration—a few criteria recover most of M_i , and no finite rubric recovers all ($\alpha_i \approx 1 \Rightarrow$ no short faithful rubric).

The recapture machinery (three jobs; a generator is not a certifier). The census is a multiple-recapture design: strong LMs, data slices, and Ω -discovery algorithms are independent *lists*, each capturing a subset of $\mathcal{B}_E$; the cell captured by none is the missing impact. Keep three jobs distinct—bound missing *mass* (Good–Turing f_1/n , needs a *stationary* proposal), convert mass to *impact* (value-Lipschitz / EVT), extend *support* past one proposal (multi-list overlap)—with two cautions: the top (k -way) interaction is *non-identifiable* (Fienberg: a 2^k contingency table with one missing cell), so source *diversity* pushes the irreducible “= 0” assumption to higher order rather than removing it; and every realistic correlation is *positive* (shared pretraining), inflating overlap into *false saturation*, so report the max-dependence-corrected lower bound, not the optimistic point estimate. This forces *two separate streams*: a frozen breadth stream (multi-family, novelty-tilted, iid—the only valid input to the mass estimator) and an adaptive depth stream (GEPA, value-tilted, magic-word hunter). **GEPA is a generator, not a certifier:** $R \leq T(m_\omega)$ immunizes the *optimality* claim against any exploration bias (a biased prompt with $R < T$ is simply *detected*, never a false optimum), but GEPA’s adaptive single-family draws break Good–Turing, so it is disqualified from the breadth/mass estimator. Net: a ladder of conditional global claims, each buying tightness with one more assumption—DPI alone gives $\text{OPT} \leq T$; + stationarity gives ε -optimality relative to the proposal support; + smoothness gives an impact bound $\text{OPT}_{\text{supp}} \leq \text{OPT}_\Omega + \hat{\Delta}$; + k independent sources extends the support and shrinks $\hat{\Delta}$ geometrically, off the zero-mass tail where only DPI guards.

Consensus stabilization c_∞ (the constructive unsupervised route). The only ground truth without labels is self-consistency among reachable criteria. Draw criteria coherent with the cluster from $\mathcal{B}_E$ (beyond Ω), weight each by its *reconstructability* $R(s)$ (its §1 recovery—this keeps the *generalizable* agreement and strips shared-but-non-generalizable artifacts; the value target is R , not the verdict), and track a *cross-source* consensus as the pool grows:

$$c_k = \text{top } R\text{-weighted factor of proposer-family } k\text{'s columns}, \quad c_\infty = \text{agreement across families } \{c_k\},$$

so per-family biases cancel (the independence that matters here is *source*-independence, not the coverage-iid of the Good–Turing certificate—a different requirement, do not conflate). Split-half alignment $|\cos(c_A, c_B)| \rightarrow 1$ pins the ideal; a plateau < 1 means no stable ideal (incoherent cluster).

The reported ceiling is $H(c_\infty)$, a Good–Toulmin-style extrapolation of the stabilization curve: assumption-bounded (smoothness/saturation), a *point estimate*, not a certificate. Its one irreducible caveat is **not** the Y problem (there is no Y): c_∞ is M_i^* *up to shared proposer bias*, and the only unsupervised lever that widens past it is genuine proposer-support diversity (different model families). Note c_∞ is distinct from α : the criterion *count* $\alpha_i \approx 1$ by proposer creativity even while the consensus converges—so measure the *stabilization rate*, not α .

The prompt-free spectral ceiling OPT' . Work *backwards* from recovery: $R(m)$ is defined for *any* labeling m , so

$$\text{OPT}' := \max_{m \in \{0,1\}^N} R(m) \geq R^* = \max_{s \in \mathcal{B}_E} R(s), \quad R^* \leq \text{OPT}' \leq \text{cap}_f,$$

a prompt-free upper bound between R^* and the cap. High- R labelings live in the decoder’s learnable subspace, so it is computed *spectrally*—PCA / kernel-PCA on a frozen embedding $\varphi(X)$, threshold the top- k directions, evaluate R —in $O(Nd^2)$, never 2^N . It is the decoder’s *capacity* (the easiest-to-decode function of X , typically a trivial surface feature such as length or topic), hence *metric-agnostic* and conservative: it over-bounds every metric at once, and is *not* $I(X; M_i^*)$. It is useful (strictly above R^*) only with a decoder *stronger* than the prompt class (a trained probe on full embeddings); with a same-class decoder $\text{OPT}' \approx R^*$. Distinct from c_∞ : same spectral machinery, but OPT' is raw learnability (metric-agnostic ceiling) while c_∞ is the criteria’s agreement (an estimate of the ideal); read $\text{headroom}' = \text{OPT}' - \hat{R}$ as a floor on how much room *could* exist, never as the operative certificate (that remains $T(m_\omega) - R$).

Process-relative scope. We never sample $\mathcal{B}_E$ iid: every generator (LLM free-gen, GEPA, curated children) is a non-IID biased process. So read every $\mathcal{B}_E$ claim—coverage, $H(c_\infty)$, “the ceiling on the ideal”—as scoped to *what the specific generator set reaches*, never $\mathcal{B}_E$ in the absolute. This is the right scope, not a defect: the generators are *diverse*, so cross-process agreement cancels their idiosyncratic biases while their *common* bias is a named, non-removable floor; and the scope is replicable and resource-invariant (it bounds what anyone using the same generators achieves at *any* compute). Our methods certify a property of exploration by a specific, diverse, non-IID set of criterion-generating processes—not a property of $\mathcal{B}_E$ itself. State the process-set; report the claim relative to it.

13 Related work

MI prompt selection (Sorensen et al. 2022). Selecting a prompt to maximize $I(\text{input}; \text{output})$ is exactly maximizing our transmission $T(p) = I(\text{verdict}; X)$; that line validates T as the right label-free signal. Our addition is to turn selection into a two-sided *certificate*— T is not merely a score to climb but the attained recovery ceiling (§3), bracketed above by the $\mathcal{B}_E$ census (§12).

Exhaustive-search prompt optimality (Wenliang et al. 2026). Their optimum—the best conditioning sequence for a near-Bayes-optimal predictor—is found by exhaustive search: exact but exponential, and established empirically rather than proved. Our $\mathcal{B}_E$ census is the cheaper, scalable counterpart: a process-relative point estimate (polynomial, not a certificate) in place of an exact exponential search. Their Bayesian framing also *names* our posterior readout $\eta(X) = \mathbb{E}[m_\omega | X]$ as the Bayes-optimal object that attains T —a lineage note (Solomonoff/Hutter), not a change to the optimum, and one that touches only the lower (T) side, never the $\mathcal{B}_E$ upper bound.

Others. Picca (2025) frames the executor as a semiotic channel—a capacity view consonant with our cap (§4); the $L^2\text{M}$ MI power-law is a scaling result adjacent to our discovery-scaling law

(§9), and Robertson–Koyejo quantify step-by-step information. None of these gives the label-free, two-sided $(R/\mathcal{B}_E)$ bracket.

14 Limitations

1. **Pure R .** Every bound certifies $R = \text{I}_{\text{TVD}}(M_{\text{te}}; \widehat{M}_{\text{te}})$, not a scalarized production objective (transfer needs component-wise convexity, the blend’s γ , and budget caps as a knapsack).
2. **Single executor.** The optimum is E -specific; cross-executor aggregation (average / min / consensus target) changes the certificate—indeed the c_∞ construction (§12) is exactly a principled cross-*source* aggregation on the ideal side.
3. **No Y , and $R \neq$ correctness.** The certificate is for recovery of M_i , not for any aggregate outcome $Y = f(M_1, \dots, M_j)$: there is *no* Y in the loop, and the ideal M_i^* is a meaning object, not a label. *If* one had Y , the wall $\text{I}(Y; \widehat{M}) \leq \min(R, \text{I}(M_i; Y))$ would still hold—but that is out of scope; the ideal is approached unsupervised (§12), never via a label.
4. **Coverage.** Rung 2 is within Ω ; closing to the global optimum on the operationalized side is free from $T(m_\omega)$ (§3), and on the *ideal* side is the census of §12, certified only under measured tail / smoothness conditions.
5. **Process-relative $\mathcal{B}_E$.** We never sample $\mathcal{B}_E$ iid; all coverage / $H(c_\infty)$ / “ceiling on the ideal” claims are scoped to a specific non-IID set of generators {LLM, GEPA, children} and are floored by their shared bias (§12). Replicable and resource-invariant, but not absolute.
6. **Estimation.** f-MIs are finite-sample (bootstrap CIs); high-order joint MIs are demanding.
7. **Validity is the load-bearing risk, not submodularity.** I_{TVD} is correlation, not causal; a side-channel (length/format) can inflate R . We block channels by construction (v-information predictor-class + adversarial saturation + counterfactual validity, Rem. 2), not by causal adjustment. Submodularity (§7) is an optional top layer (the large- Ω approximation + tail-bound); the core—cap, DPI, the recovery ceiling $T(m_\omega)$, exact $|\Omega| \leq 15$ certificate, A —is submodularity-free, so the theory degrades gracefully (measured optimum + valid ceiling, no certified approximation ratio) if γ cannot be established.